

Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 5,000 - Columns: 18 - Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Customer Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
1	56	Female	Skirt	Footwear	15	New York	XL	Blue	Spring	4	Yes	Free Shipping	Yes	No	29	PayPal
2	69	Male	Skirt	Footwear	21	Phoenix	XXL	Green	Fall	2	No	Next Day Air	Yes	Yes	24	Bank Transfer
3	46	Female	Jacket	Accessories	14	Los Angeles	L	Brown	Summer	3	No	Standard	Yes	Yes	39	Credit Card
4	32	Male	Jeans	Clothing	64	Chicago	M	Brown	Spring	2	Yes	Free Shipping	No	No	34	Bank Transfer
5	60	Male	Blouse	Clothing	127	Los Angeles	XL	White	Fall	3	No	Next Day Air	Yes	No	36	Cash

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created age_group column by binning customer ages.
 - Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	640868
2	Male	631501

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint		purchase_amount bigint
1		7	494
2		11	430
3		12	342
4		15	360
5		16	367
6		17	337
7		19	436
8		22	324
9		25	496
10		28	437

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	lock	Average Product Rating numeric	lock
1	Coat		3.10	
2	Jeans		3.07	
3	Skirt		3.03	
4	Sweater		3.02	
5	Jacket		3.01	

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	lock	round numeric	lock
1	Standard		253.07	
2	Express		254.83	

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	2454	254.93	625589.00
2	No	2546	254.04	646780.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	lock	discount_rate numeric	lock
1	Sneakers		54.15	
2	Jacket		53.06	
3	Shorts		52.53	
4	Coat		52.24	
5	T-Shirt		51.98	

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	lock	Number of Customers bigint	lock
1	Loyal		4014	
2	New		88	
3	Returning		898	

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	Item_rank bigint	category text	Item_purchased text	total_orders bigint
1	1	Accessories	Coat	146
2	2	Accessories	T-Shirt	139
3	3	Accessories	Shoes	135
4	1	Clothing	Skirt	137
5	2	Clothing	Handbag	136
6	3	Clothing	Coat	135
7	1	Footwear	Shorts	143
8	2	Footwear	Jacket	135
9	3	Footwear	Jeans	125

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2237
2	Yes	2168

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	326182
2	Middle-aged	325377
3	Adult	316695
4	Senior	304115

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.